

Discussion Module:

Spatial (not geospatial),
Time Series

Learn to deliver DSE spatial queries, including time series. For example:

- Did the pizza driver pass a given point ? (Where's my pizza ?)
- Is there hotel room availability for a given date ?

Prerequisites to this Discussion Module:

- That you are comfortable with making and using DSE Search geospatial objects and queries.

Discussion Lab:

Open Probe

Next page: Describe/offer business use cases-



Discuss: Spatial, and Time-



Describe business use cases for:

- Knowing that a driver passed through a given location ?
- Or didn't pass through a given location ?
- Is in a location ?
- A resource is/is-not available on a given date ?

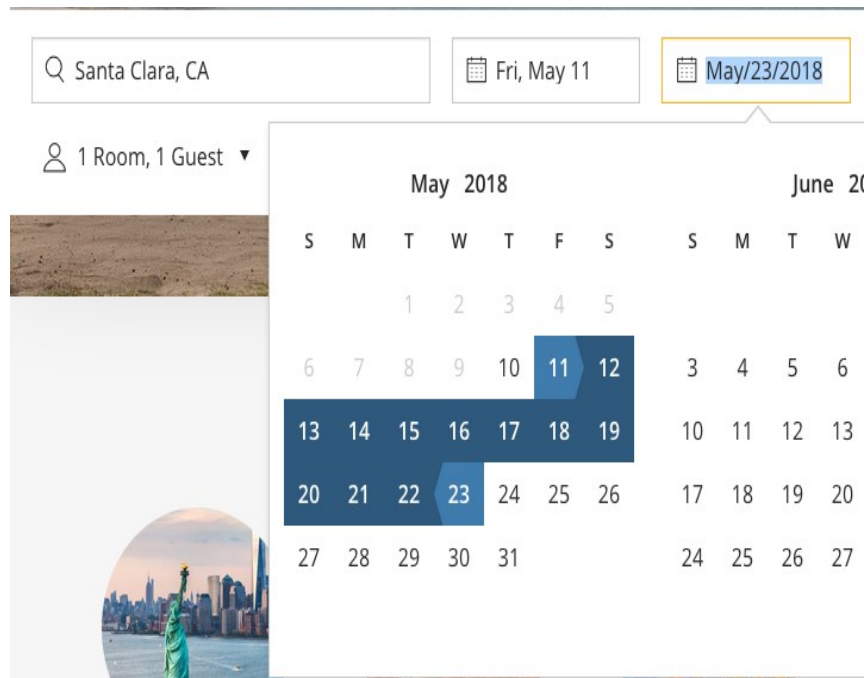
How is a Lat/Lon pair similar to a span of time ?

End of Discussion Lab:



DSE Search, Spatial, Time Series

- Use RPT, geo='false'
- Set; worldBounds, maxDistErr, distErrPct
- Casting; POINT, **LINESTRING**
- Operators; IsWithin, Intersects, [] (Range)



End to End Example:

IsWithin, Intersects, POINT,
LINESTRING

Working with routes-



IsWithin, Intersects, CREATE KEYSPACE



```
DROP KEYSPACE IF EXISTS ks_7455;
```

```
CREATE KEYSPACE ks_7455 WITH  
REPLICATION =
```

```
{'class': 'SimpleStrategy',  
  'replication_factor': 1};
```

```
USE ks_7455;
```

IsWithin, Intersects, CREATE TABLE

```
CREATE TABLE t1
```

```
(  
  col1          TEXT,  
  my_point      'PointType',  
  my_line       'LineStringType',  
  PRIMARY KEY ((col1))  
);
```



Or TEXT-
There really aren't
sane/usable assumptions
for these, to allow default
generation

OLD
OLD
OLD

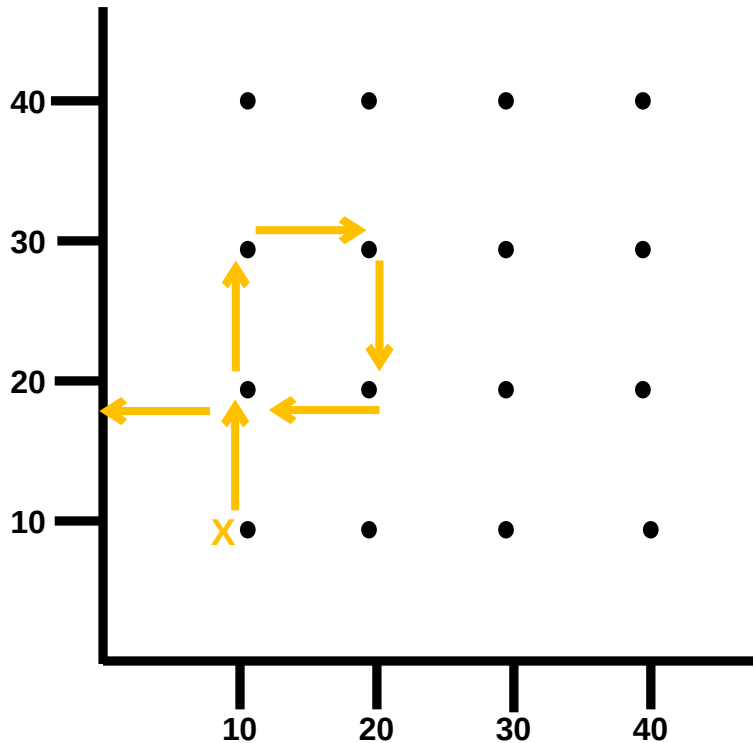
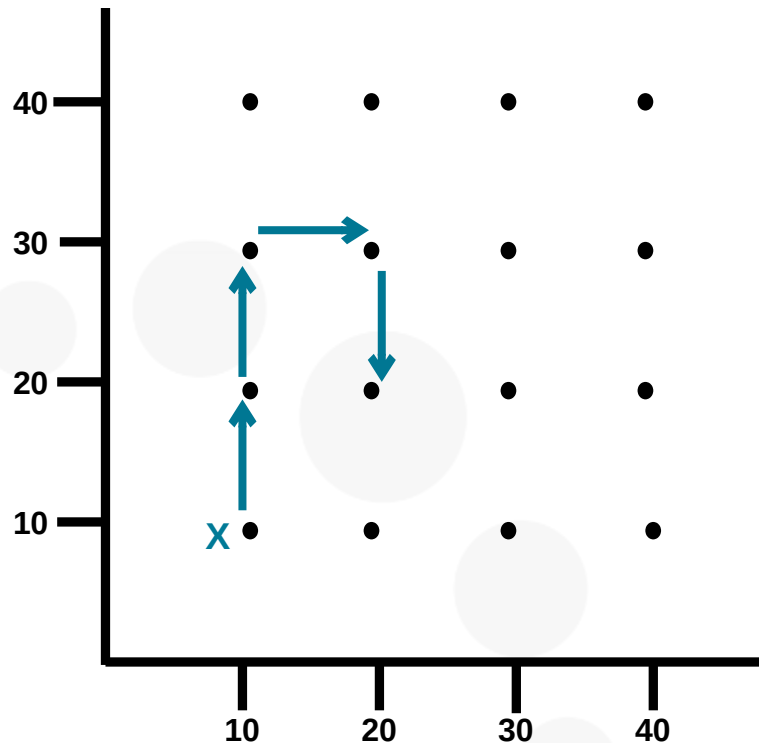


IsWithin, Intersects, Add Data

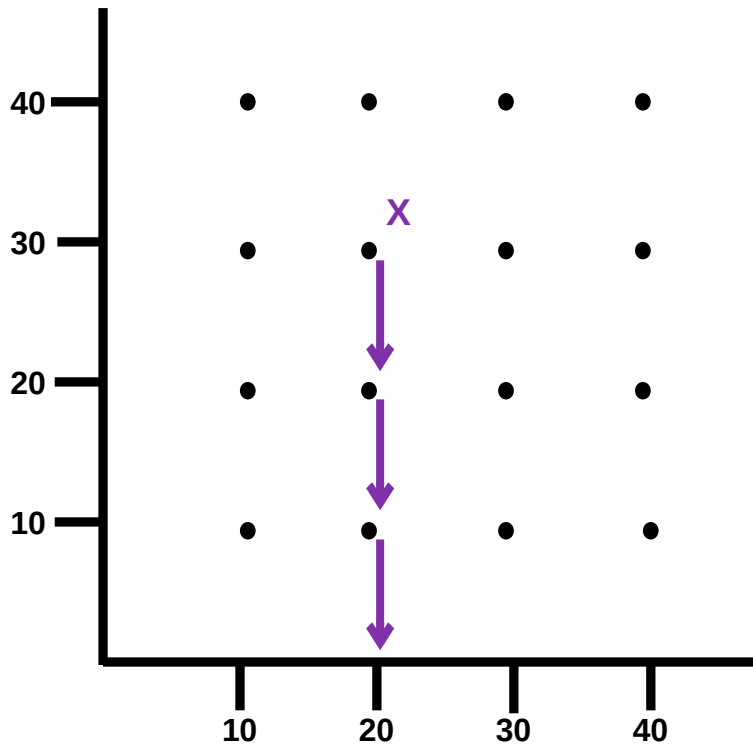
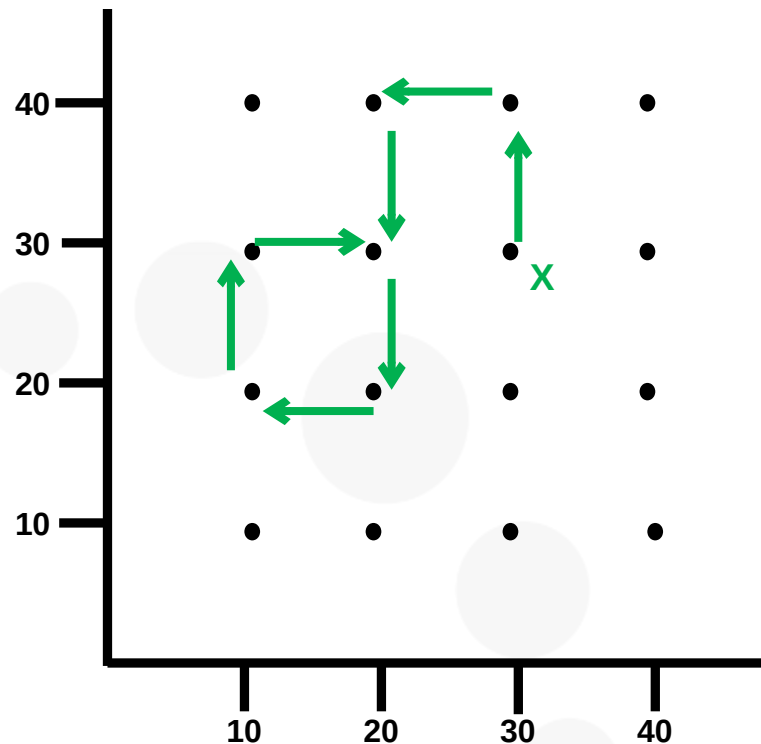
```
INSERT INTO t1 (col1, my_point, my_line)
VALUES ('1', 'POINT(10 10)',
        'LINESTRING(10 10, 10 20, 10 30, 20 30, 20 20)' );

VALUES ('2', 'POINT(10 10)',
        'LINESTRING(10 10, 10 20, 10 30, 20 30, 20 20, 20 10, 20 0)' );
VALUES ('3', 'POINT(30 30)',
        'LINESTRING(30 30, 30 40, 20 40, 20 30, 20 20, 10 20, 10 30, 20 30)' );
VALUES ('4', 'POINT(20 30)',
        'LINESTRING(20 30, 20 20, 10 20, 0 20)' );
```

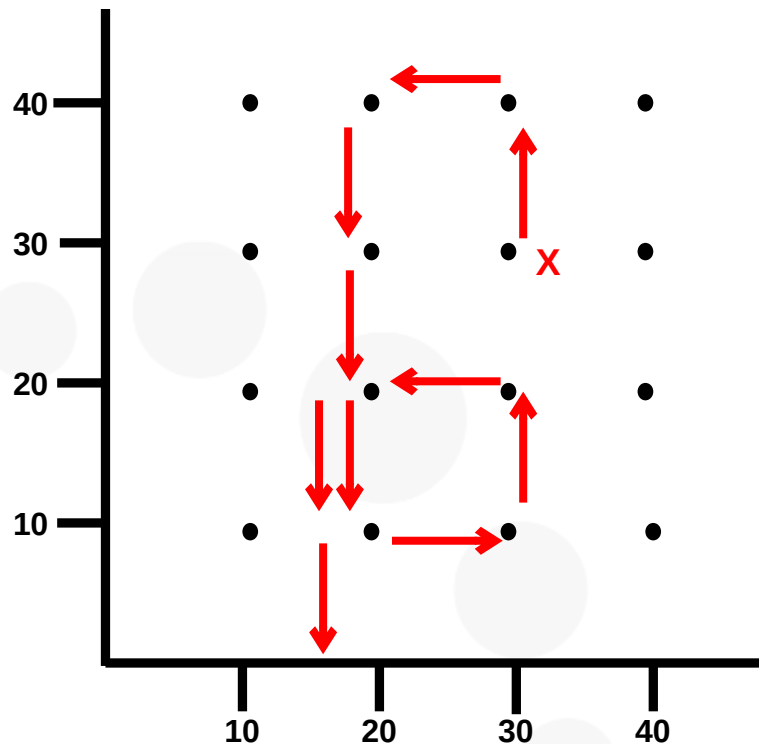
Same Data, Plotted, Page 1 of 2



Same Data, Plotted, Page 2 of 2



LINESTRING data can not-



InvalidRequest: Error from server: code=2200 [Invalid query]
message="Unable to make LineStringType from
'LINESTRING(30 30, 30 40, 20 40, 20 30, 20 20, 20 10, 30 10,
30 20, 20 20, 20 10, 20 0)' 'LINESTRING (30 30, 30 40, 20 40,
20 30, 20 20, 20 10, 30 10, 30 20, 20 20, 20 10, 20 0)' is not
simple. Points and edges cannot self-intersect."

JIRA, DSP-11516

You can not *smudge* the values. E.g., add 0.00001 or similar to overlapping points.

<https://datastax.jira.com/browse/DSP-11516>

You have to break these LINESTRINGS in two-

**DataStax
Internal Use
Only**

IsWithin, Intersects, CREATE SEARCH INDEX



```
// CREATE SEARCH INDEX ON t1
//   WITH COLUMNS col1, my_point, my_line;
//
// InvalidRequest: Error from server: code=2200 [Invalid query]
// message="The search index schema could not be auto-generated
// because: Type org.apache.cassandra.db.marshall.LineStringType
// is not supported with automatic Solr schema generation.
// Specify 'lenient: true' in your resource generation options
// to ignore unsupported columns."
```

```
CREATE SEARCH INDEX ON t1
  WITH COLUMNS col1, * {excluded : true } ;
```


ALTER SEARCH INDEX, deploy

```
ALTER SEARCH INDEX SCHEMA ON t1
ADD types.fieldType[@name='ft_rpt',
@class='solr.SpatialRecursivePrefixTreeFieldType',
@geo='false',
@worldBounds='ENVELOPE(-1000, 1000, 1000, -1000)',
@maxDistErr='0.001', @distanceUnits='degrees'];
```

```
ALTER SEARCH INDEX SCHEMA ON t1
ADD fields.field[@name='my_point', @type='ft_rpt',
@indexed='true', @multiValued='false', @stored='true',
@docValues='true'];
```

```
ALTER SEARCH INDEX SCHEMA ON t1
ADD fields.field[@name='my_line', @type='ft_rpt',
@indexed='true', @multiValued='false', @stored='true',
@docValues='true'];
```

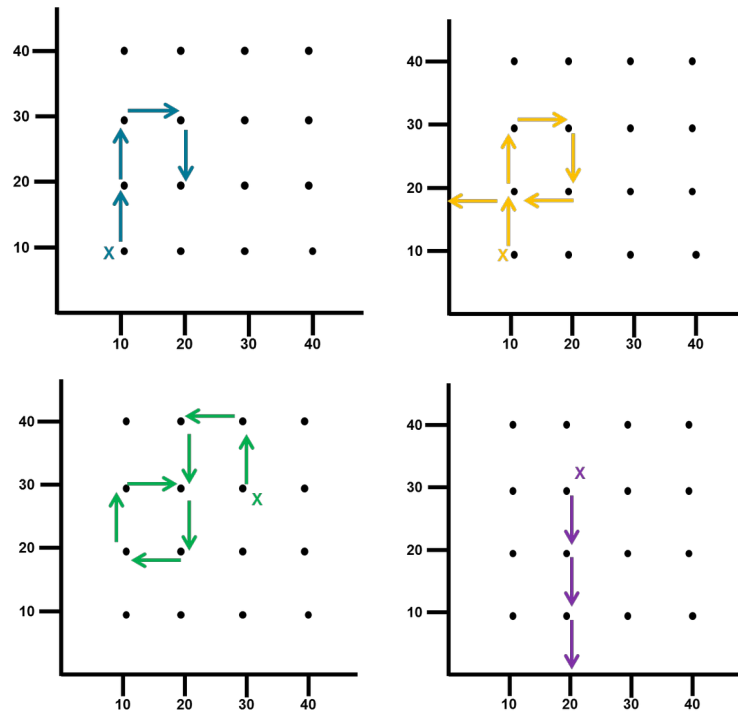
```
RELOAD SEARCH INDEX on t1;
REBUILD SEARCH INDEX ON t1;
```



SELECT, IsWithin POINT

```
SELECT col1, my_point FROM t1
WHERE solr_query=
  '{ "q" : "my_point : \"IsWithin(BUFFER(
    POINT(30 30), 1))\""}';
```

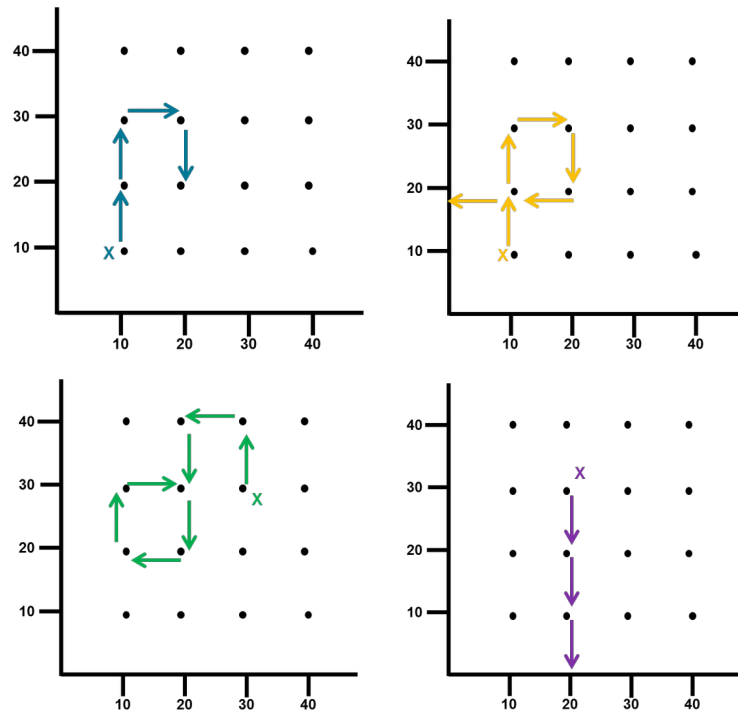
col1	my_point
3	POINT (30.0 30.0)



SELECT, IsWithin POINT

```
SELECT col1, my_point FROM t1
WHERE solr_query=
  '{ "q" : "my_point : \"IsWithin(BUFFER(
    POINT(20 30), 1))\""}';
```

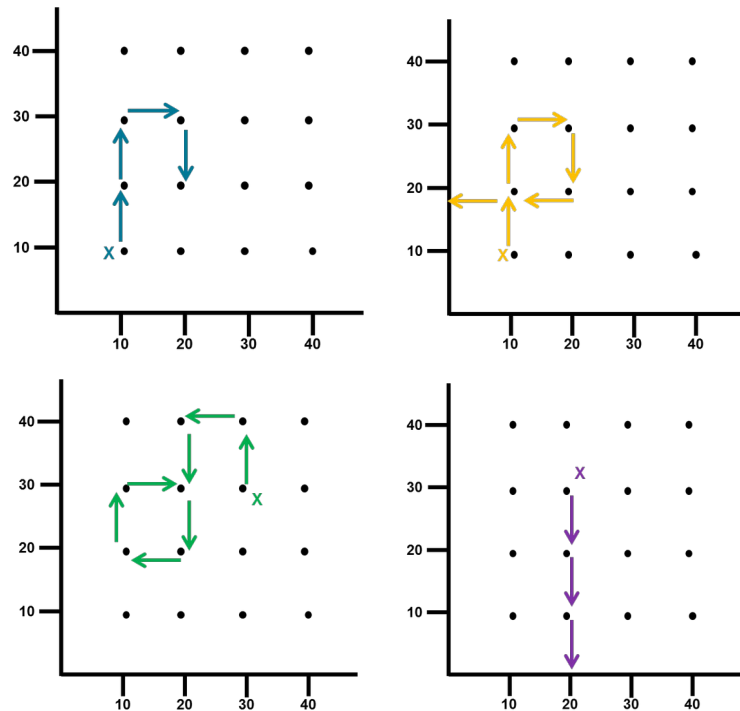
col1	my_point
4	POINT (20.0 30.0)



SELECT, IsWithin POINT

```
SELECT col1, my_point FROM t1
WHERE solr_query=
  '{ "q" : "my_point : \'IsWithin(BUFFER(
    POINT(20 30), 10))\'"' }';
```

col1	my_point
4	POINT (20.0 30.0)
3	POINT (30.0 30.0)



IsWithin using LINESTRING, Odd results-

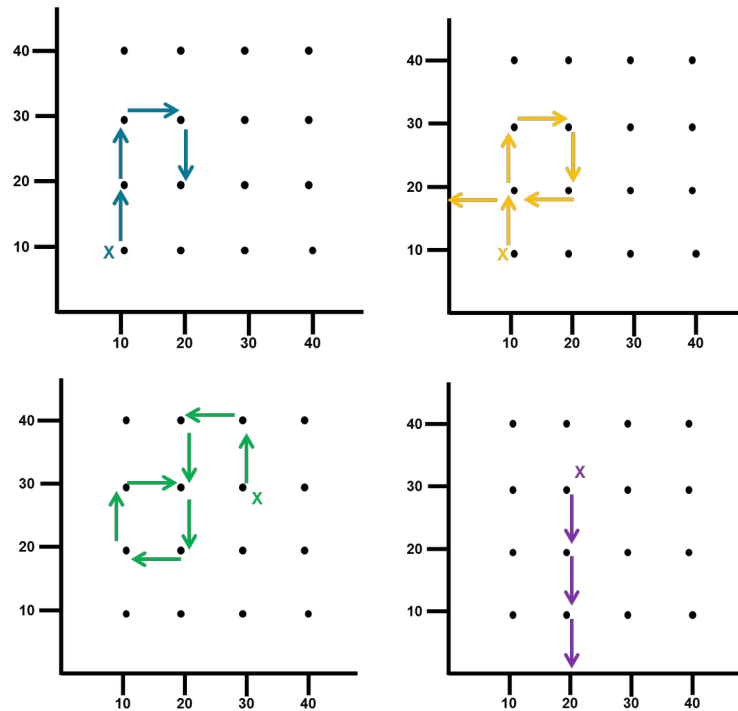
```
SELECT * FROM t1 WHERE  
  solr_query=  
'{ "q" : "my_line : \"IsWithin(BUFFER(  
  POINT(20 30), 40))\""}';
```



DataStax
Internal Use
Only

LINESTRING, Intersects

```
SELECT col1, my_line
FROM t1 WHERE
solr_query='my_line:"Intersects(
POINT(20 20))"';
```



col1	my_line
4	LINESTRING (20.0 30.0, 20.0 20.0, 10.0 20.0, 0.0 20.0)
3	LINESTRING (30.0 30.0, 30.0 40.0, 20.0 40.0, 20.0 30.0, 20.0 20.0, 10.0 20.0, 10.0 30.0, 20.0 30.0)
2	LINESTRING (10.0 10.0, 10.0 20.0, 10.0 30.0, 20.0 30.0, 20.0 20.0, 20.0 10.0, 20.0 0.0)
1	LINESTRING (10.0 10.0, 10.0 20.0, 10.0 30.0, 20.0 30.0, 20.0 20.0)

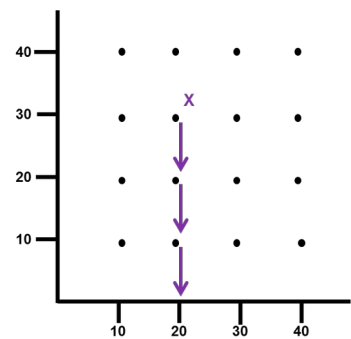
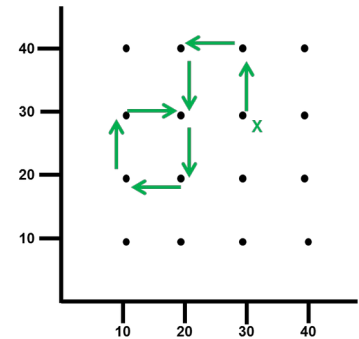
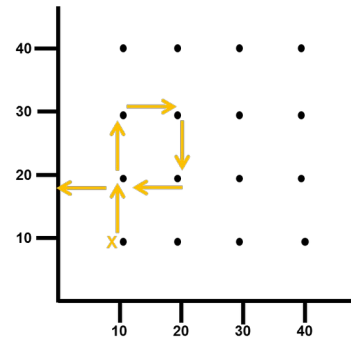
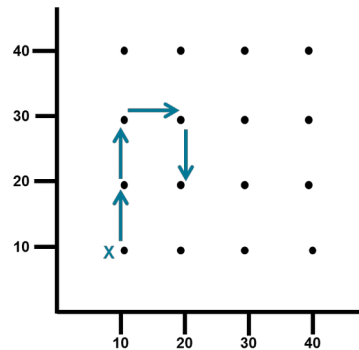
LINESTRING, Intersects

```
SELECT col1, my_line FROM t1
WHERE solr_query='my_line:
  "Intersects(POINT(30 40))";
```

```
SELECT col1, my_line FROM t1
WHERE solr_query=
  '{ "q" : "my_line : \"Intersects(
  POINT(30 40))\""' }';
```

```
// same for both
col1 | my_line
```

```
3 | LINESTRING (30.0 30.0, 30.0 40.0, 20.0 40.0, 20.0 30.0, 20.0 20.0, 10.0 20.0, 10.0 30.0, 20.0 30.0)
```



Working with routes-

End to End Example:

IsWithin, Intersects, POINT,
LINESTRING
REVIEW



End to End Example:

Ranges of dates/times.

Working with time series-



```
CREATE TABLE t2
(
  col1          TEXT,
  my_dates      'PointType',
  PRIMARY KEY ((col1))
);
```

```
INSERT INTO t2 (col1, my_dates)
VALUES ('101', 'POINT(2 5)');
```

```
VALUES ('102', 'POINT(2 3)');
```

```
VALUES ('103', 'POINT(4 5)');
```

```
VALUES ('104', 'POINT(1 2)');
```

```
VALUES ('105', 'POINT(1 3)');
```

```
VALUES ('106', 'POINT(1 6)');
```

```
CREATE SEARCH INDEX ON t2
WITH COLUMNS col1, * {excluded : true } ;
```

CREATE TABLE, add data, CREATE SEARCH INDEX



ALTER SEARCH INDEX

```
ALTER SEARCH INDEX SCHEMA ON t2
ADD types.fieldType[@name='ft_rpt',
@class='solr.SpatialRecursivePrefixTreeFieldType',
@geo='false',
@worldBounds='0 0 365 365',
@maxDistErr='1', @distErrPct='0'];
```



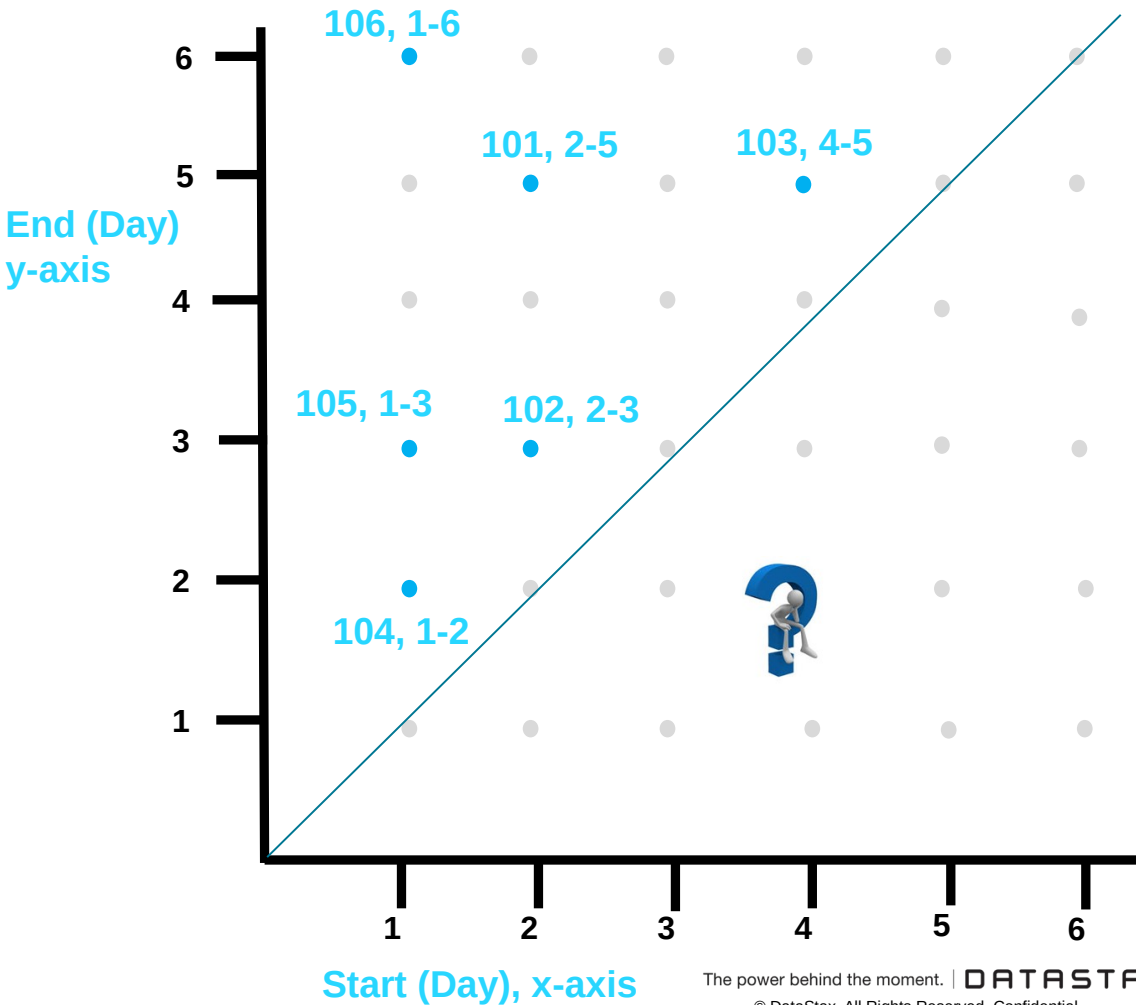
No ENVELOPE, like earlier



```
ALTER SEARCH INDEX SCHEMA ON t2
ADD fields.field[@name='my_dates', @type='ft_rpt',
@indexed='true', @multiValued='false', @stored='true',
@docValues='true'];
```

```
RELOAD SEARCH INDEX on t2;
REBUILD SEARCH INDEX on t2;
```

Same Data, Plotted

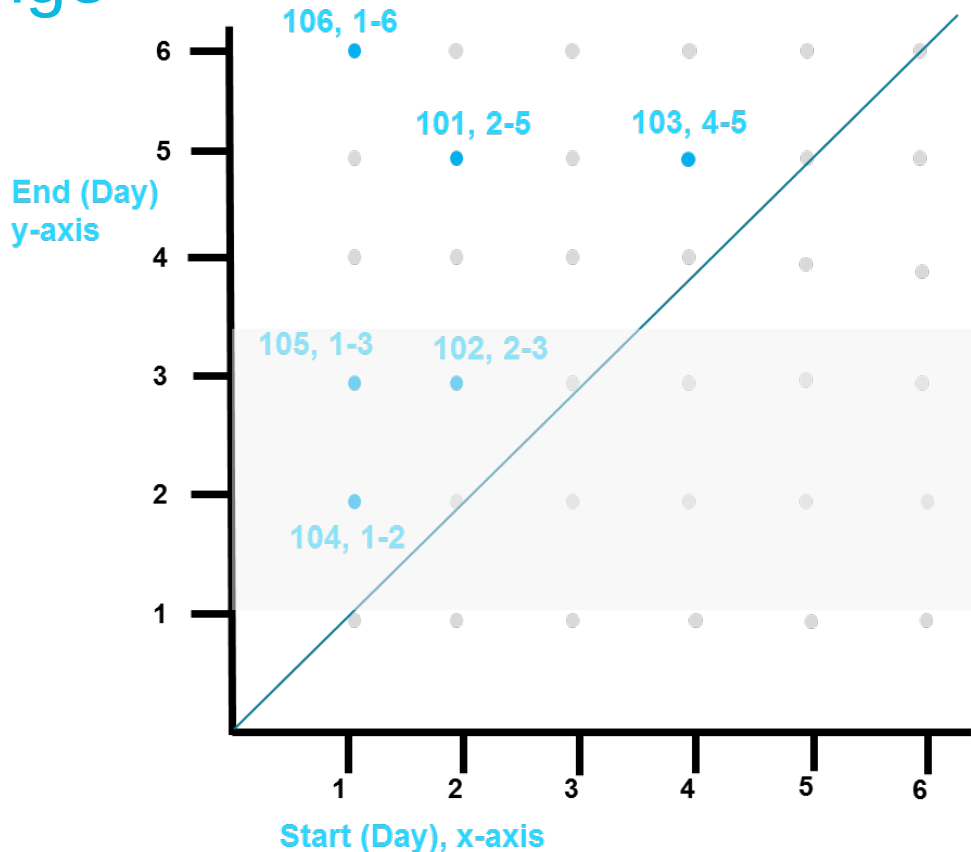


Contains; exclusive to range

```
SELECT col1, my_dates
FROM t2
WHERE solr_query=
  '{ "q" : "*:~*", "fq" : "my_dates:
    [0,1 TO 3,365]" }';
```

↑ x ↑ y ↑ y ↑ x

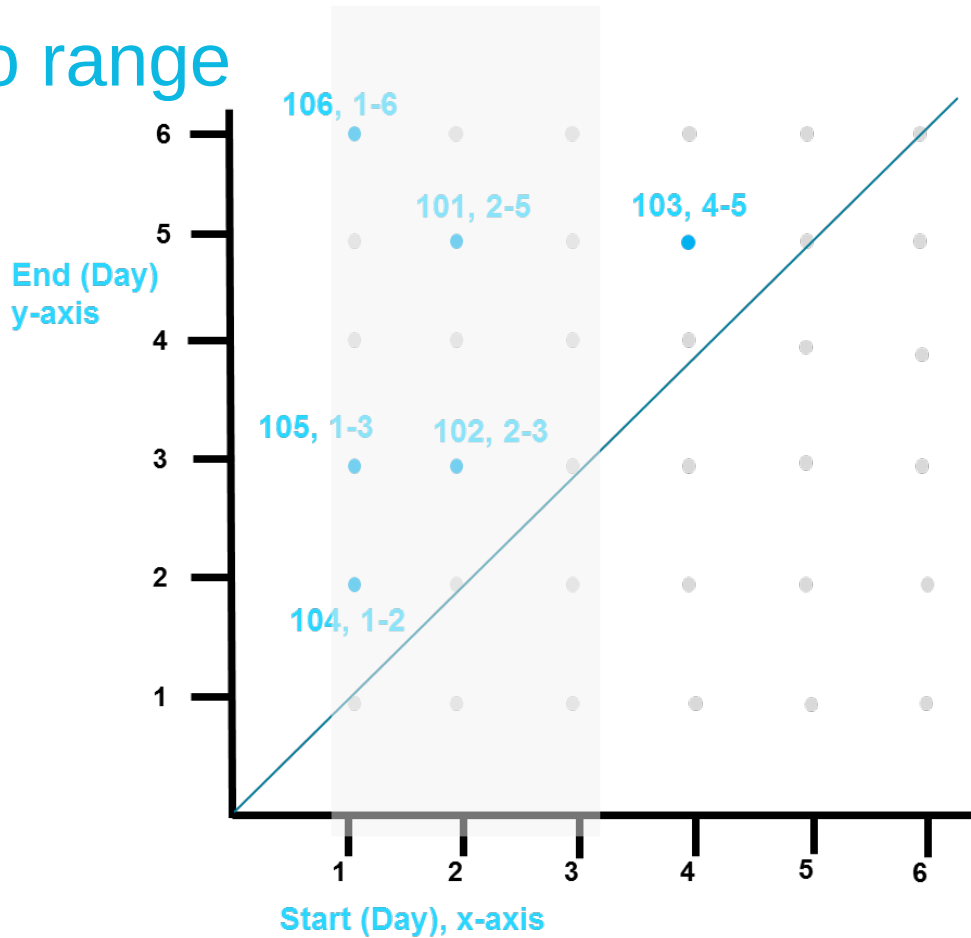
col1	my_dates
104	POINT (1.0 2.0)
102	POINT (2.0 3.0)
105	POINT (1.0 3.0)



Intersects; not exclusive to range

```
SELECT col1, my_dates
FROM t2
WHERE solr_query=
  '{ "q" : "*:~*", "fq" : "my_dates:
    [1,0 TO 365,3]" }';
```

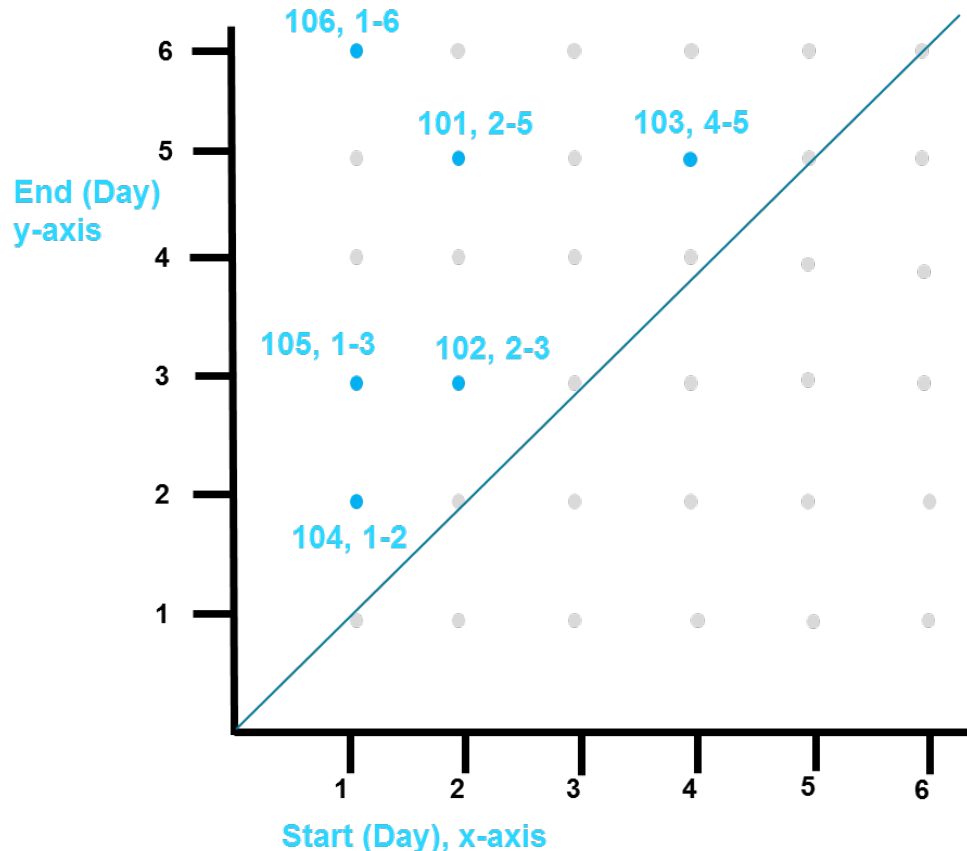
col1	my_dates
104	POINT (1.0 2.0)
106	POINT (1.0 6.0)
102	POINT (2.0 3.0)
105	POINT (1.0 3.0)
101	POINT (2.0 5.0)



On a given date/time

```
SELECT col1, my_dates
FROM t2
WHERE solr_query=
  '{ "q" : "*:*",
    "fq" : "my_dates:
    [4,0 TO 10,4]" }';
```

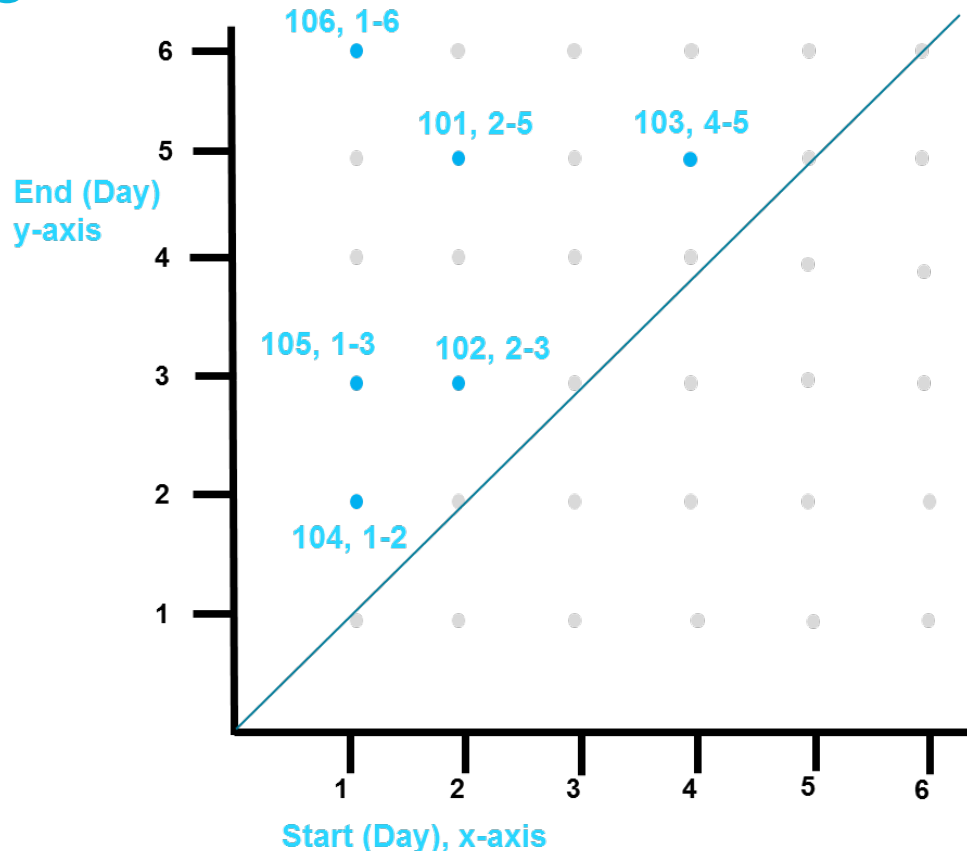
col1	my_dates
103	POINT (4.0 5.0)
106	POINT (1.0 6.0)
101	POINT (2.0 5.0)



Not on a given date/time

```
SELECT col1, my_dates
FROM t2
WHERE solr_query=
  '{ "q" : "*:*",
    "fq" : "NOT
    my_dates:[4,0 TO 10,4]" }';
```

col1	my_dates
104	POINT (1.0 2.0)
102	POINT (2.0 3.0)
105	POINT (1.0 3.0)



End to End Example:

Ranges of dates/times.

REVIEW

Working with time series-





End of Module: